

# THE MECHANICS OF A BLOW-UP

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**ABSTRACT.** This study note establishes the foundational geometric and algebraic machinery of the blowup construction, serving as a critical prerequisite for understanding the exceptional curves, resolution graphs, and Stein fillings discussed by Plamenevskaya and Starkson. We first formalize the classical construction of the blowup of  $\mathbb{C}^n$  at the origin, explicitly identifying the exceptional divisor  $E \cong \mathbb{CP}^{n-1}$  and analyzing the behavior of total and proper transforms.

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## 1. BLOW-UPS

**Definition 1.1.** Let  $S$  be a surface and  $p \in S$ . Then there exist a surface  $\hat{S}$  and a morphism  $\epsilon : \hat{S} \rightarrow S$ , which are unique up to isomorphism, such that

- (i) the restriction of  $\epsilon$  to  $\epsilon^{-1}(S - \{p\})$  is an isomorphism onto  $S - \{p\}$ ;
- (ii)  $\epsilon^{-1}(p) = E$ , say, is isomorphic to  $\mathbb{P}^1$ .

To note, recall that an isomorphism between two complex space germs is the topological and algebraic equivalent of saying two spaces look completely identical when you zoom in infinitely close to the base points  $x$  and  $y$ . Let us denote our objects now as *complex surface germs*. We will continue to denote morphisms as *holomorphic map germs*. From definition 1.1, if there is a point  $p$  on a surface  $S$ , there are a subset of "equal" elements that form the surface  $\hat{S}$  by the morphism  $\epsilon : \hat{S} \rightarrow S$ . For this to be true, the two conditions above must hold. For the purpose of the Plamenevskaya and Starkson paper, Rather than  $S$  and  $\hat{S}$ , we have:

$$S = (S, p) = (X, 0)$$
$$\hat{S} = \tilde{X}$$

Condition (i) means we restrict our morphism  $\epsilon$  to the preimage, which is the surface minus the point  $p$ . This restricted map is an isomorphism onto the surface minus that same point. In simple terms, this works like an inverse identity: outside of the point  $p$ , the map can be flipped backwards perfectly 1-to-1. The only place this identity-like matching breaks down is right at  $p$ . For purposes of the Plamenevskaya and Starkson paper,  $p$  is our singularity. Condition (ii) tells us what happens

to the exact point where our identity map broke down. If we take the preimage of just that single point  $p$ , it expands into a whole new set  $E$ . This set is isomorphic to  $\mathbb{P}^1$ , which means topologically it is just a 2-sphere. So, the blowup essentially replaces a single point on our surface with an entire sphere. To actually build the map germ  $\epsilon$ , we first choose a tiny neighborhood  $U$  around our point  $p$ . Inside this neighbourhood, we can pick local coordinates  $x$  and  $y$  that make the point  $p$  look like the origin  $(0, 0)$ . The grid lines  $x = 0$  and  $y = 0$  cross each other like a standard flat coordinate axis, meaning we are basically just treating a small patch of our surface like a flat complex plane. Note that we also assume that the point  $p$  is the only point of intersections between the two curves. The subvariety  $\hat{U}$ , which is a smaller space located in  $U$ , is defined as a product space. This means it is the set of all possible ordered pairs containing one point from  $U$  and one point from  $\mathbb{P}^1$ :

$$U \times \mathbb{P}^1 = \{((x, y), [X : Y]) \mid (x, y) \in U \text{ and } [X : Y] \in \mathbb{P}^1\}$$

$X$  and  $Y$  are coordinates in  $\mathbb{P}^1$ . We define  $\hat{U}$  by the following equation:

$$xY - yX = 0$$

If we rearrange the equation  $xY - yX = 0$  by moving one term over, we get:

$$xY = yX$$

If we divide both sides by  $x$  and  $X$  (assuming they aren't zero), it turns into:

$$\frac{Y}{X} = \frac{y}{x}$$

This tells you that the direction you are pointing on the sphere ( $\frac{Y}{X}$ ) must perfectly match the slope of your coordinates on the flat plane ( $\frac{y}{x}$ ). Away from the origin (where at most one of  $x$  or  $y$  is zero), the slope equation forces exactly one unique point on the sphere for each point on the plane. Because this matching is perfectly 1-to-1, the projection map  $\epsilon$  is a flawless isomorphism outside of the origin.

Right at the origin  $p = (0, 0)$ , the equation becomes  $0 = 0$ , so every point on the sphere is valid. This means the preimage of the origin is the entire sphere, written formally as  $\{p\} \times \mathbb{P}^1$ .

To build the complete blown-up surface  $\hat{S}$ , we take the original surface minus the point,  $S \setminus \{p\}$ , and paste our new patch  $\hat{U}$  onto it. We can do this cleanly because the parts away from the origin are completely identical, leaving us with a smooth global surface where the point  $p$  has been smoothly replaced by a 2-sphere.

## 2. TASK: CALCULATION OF TRANSFORMS, AND VISUALISATION UNDER A BLOW-UP

Given the singular curve defined by the equation:

$$f(x, y) = y^3 - x^2 = 0$$

1. Write down explicit coordinate charts for the blow-up of  $\mathbb{C}^2$  at the origin  $\pi : \tilde{\mathbb{C}}^2 \rightarrow \mathbb{C}^2$ .
2. Take the singular curve  $C = y^2 - x^3 = 0 \in \mathbb{C}^2$ . Compute its total transform  $\pi^{-1}(C)$  and the proper transform  $\tilde{C}$  in both parts. Show that  $\tilde{C}$  is smooth.

## Solutions to the Blow-up Task

### 1. Coordinate Chart Derivation

The blow-up of  $\mathbb{C}^2$  at the origin replaces the origin with a projective line  $\mathbb{P}^1$  representing all lines (slopes) through the origin. This space is covered by two distinct coordinate charts mapping down to the original  $(x, y)$  plane.

- **Chart 1 ( $x$ -chart):** We choose coordinates  $(u, v)$  where  $u$  represents the original coordinate  $x$ , and  $v$  represents the slope ratio  $\frac{y}{x}$ . This gives the coordinate transformation:

$$x = u, \quad y = vu$$

- **Chart 2 ( $y$ -chart):** To handle regions where  $x = 0$ , the roles flip. We choose coordinates  $(s, t)$  where  $s$  tracks the original coordinate  $y$ , and  $t$  tracks the inverse slope ratio  $\frac{x}{y}$ . This gives the coordinate transformation:

$$x = ts, \quad y = s$$

### 2. Transform Calculations

*Analysis in Chart 1.* Substituting the transformation equations  $x = u$  and  $y = vu$  into our original curve equation yields:

$$(vu)^2 - (u)^3 = 0$$

Expanding the exponents gives:

$$v^2u^2 - u^3 = 0$$

Factoring out the highest possible shared power of  $u$  (which is  $u^2$ ), we get the equation for the total transform in this chart:

$$u^2(v^2 - u) = 0$$

From this factored form, we identify the individual components:

- **Total Transform  $\pi^{-1}(C)$ :** The full algebraic equation  $u^2(v^2 - u) = 0$ .
- **Proper Transform  $\tilde{C}$ :** The remaining curve component inside the parentheses, defined by:

$$g(u, v) = v^2 - u = 0$$

*Analysis in Chart 2.* Substituting the transformation equations  $x = ts$  and  $y = s$  into our original curve equation yields:

$$(s)^2 - (ts)^3 = 0$$

Expanding the exponents gives:

$$s^2 - t^3s^3 = 0$$

Factoring out the highest possible shared power of  $s$  (which is  $s^2$ ), we get the equation for the total transform in this chart:

$$s^2(1 - t^3s) = 0$$

From this factored form, we identify the individual components:

- **Total Transform  $\pi^{-1}(C)$ :** The full algebraic equation  $s^2(1 - t^3s) = 0$ .
- **Proper Transform  $\tilde{C}$ :** The remaining curve component inside the parentheses, defined by:

$$h(s, t) = 1 - t^3s = 0$$

### 3. Smoothness via the Jacobian Criterion

A curve is smooth if there are no points on the curve where its partial derivatives vanish simultaneously.

*Jacobian Check for Chart 1.* Let  $g(u, v) = v^2 - u = 0$ . We compute the partial derivatives:

$$\frac{\partial g}{\partial u} = -1, \quad \frac{\partial g}{\partial v} = 2v$$

For a singular point to exist, both derivatives must equal zero. However,  $\frac{\partial g}{\partial u} = -1 \neq 0$  everywhere. Because the partial derivatives can never simultaneously vanish, the proper transform  $\tilde{C}$  is perfectly smooth in Chart 1.

*Jacobian Check for Chart 2.* Let  $h(s, t) = 1 - t^3s = 0$ . We compute the partial derivatives:

$$\frac{\partial h}{\partial s} = -t^3, \quad \frac{\partial h}{\partial t} = -3t^2s$$

Setting  $\frac{\partial h}{\partial s} = -t^3 = 0$  forces  $t = 0$ . If we substitute  $t = 0$  back into our proper transform curve equation  $1 - t^3s = 0$ , we get:

$$1 - (0)^3s = 0 \implies 1 = 0$$

This yields a logical contradiction. Therefore, there are no points on the curve that can satisfy the vanishing of the Jacobian criterion. The proper transform  $\tilde{C}$  is perfectly smooth in Chart 2.

### REFERENCES

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